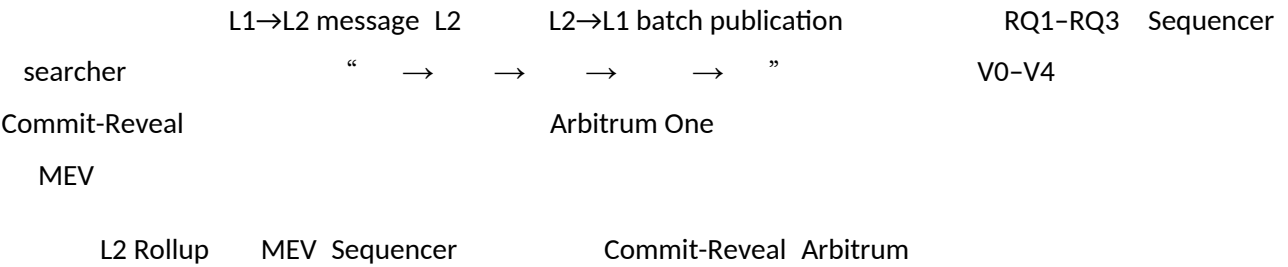
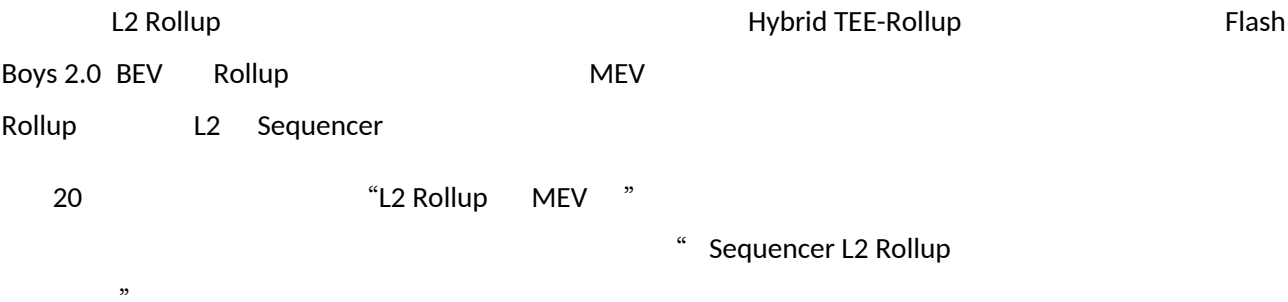


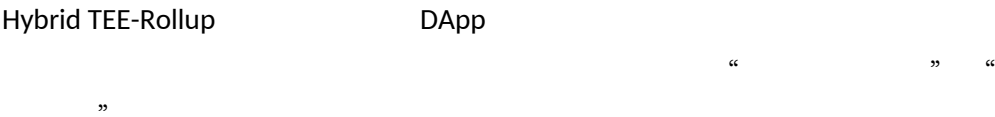
L2 Rollup MEV

L2 Rollup MEV Sequencer

2026 7 23



2.1 Hybrid TEE-Rollup



Flash Boys 2.0 Gas BEV
Rollup L1 L2 Sequencer L2

2.2 MEV L2 Sequencer

20 MEV MEV L2 DAG/
MEV BFT
Sequencer L2 Aequitas Quick Order Fairness Wendy SoK
Timeboost auction-based sequencing
MEV “

Sequencer + + ”

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

3.1

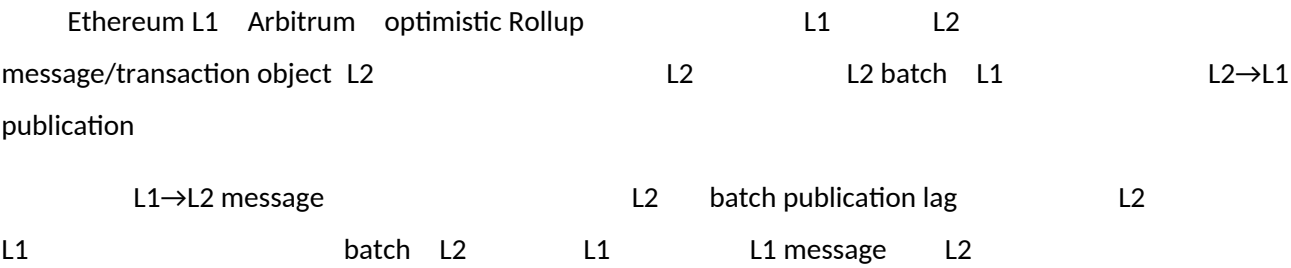
Rollup MEV 20 “
” “L2 Sequencer” MEV MEV SoK L2 MEV
Timeboost MEV “ MEV ” ordering domain
realized MEV
“ Sequencer L2 Rollup ”
L1 L2

3.2

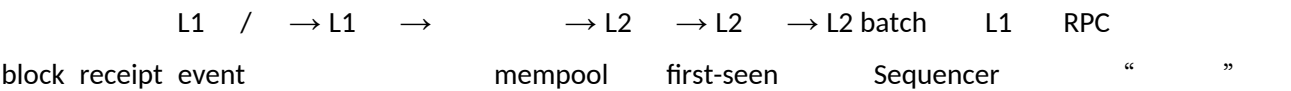
MEV SoK single/cross-domain potential/realized
Theory of MEV I
BEV ground truth L2 MEV /
“ ” “ ” Sequencer MEV
L1→L2 first-seen

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

4.1



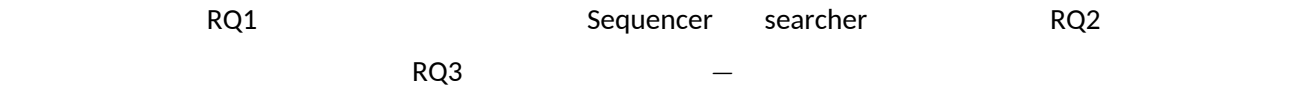
4.2



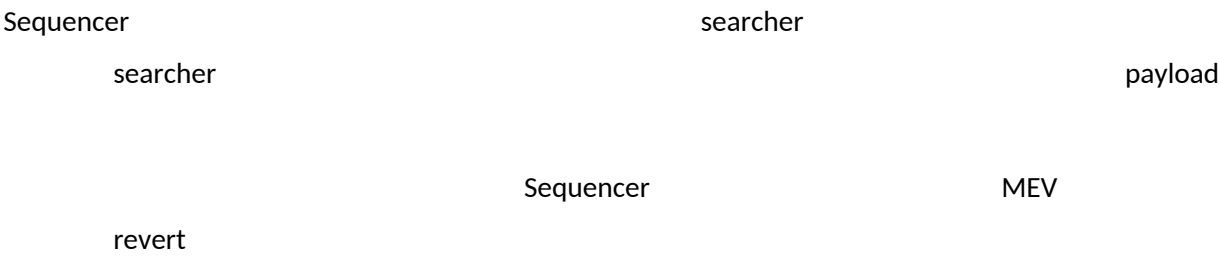
□□/□□	□□□□□□□□□□□□□□	□□□□□□□□
L1	L1 transaction event block timestamp	first-seen
		Sequencer V0
L2 /	L2 receipt event block	
L2→L1 batch	batch event	publication relation

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □

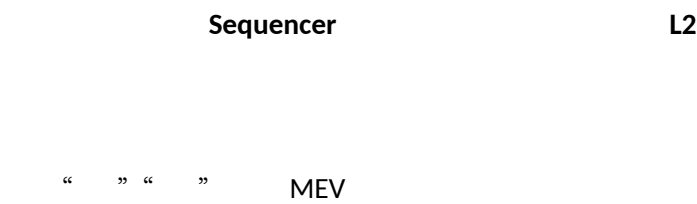
5.1 RQ1-RQ3



5.2



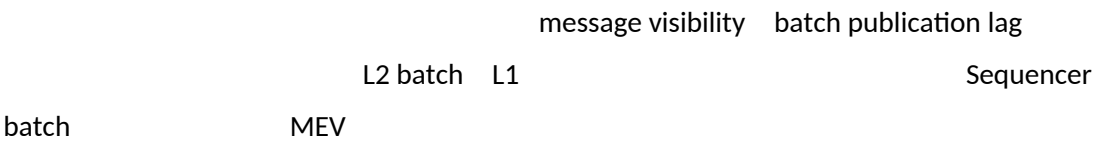
5.3



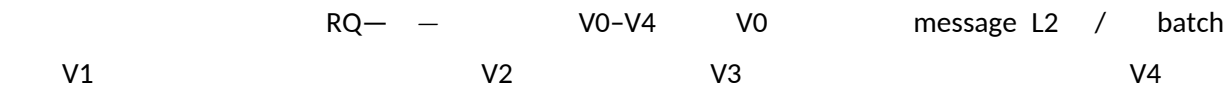
6.1



6.2



6.3



□□	□□	□□□□□□	□□□□□
V0			
V1		/	MEV
V2			
V3			
V4			MEV

6.4 Commit-Reveal

RQ3 Commit-Reveal payload “ + ”

$H(\text{domain} || \text{chainId} || \text{rollupId} || \text{sender} || \text{nonce} || \text{payload} || \text{salt})$ /rollup

nonce commitment → commitment slot → reveal payload

salt → commitment →

payload salt reveal reveal

payload Sequencer reveal

V4

6.5 Arbitrum

Arbitrum One finalized

Ethereum 40 batch 25 kind=9 retryable message batch L2 block *range*

retryable ID L2 receipt *exact* 25 retryable message exact 25/25 40

batch range 40/40 message provenance 100%

$W_{ingest} = t(L2 \text{ retryable creation block}) - t(L1 \text{ message finalized block})$ $W_{effect} = t(L2 \text{ successful redeem block}) - t(L1 \text{ message finalized block})$ $P_{lag} = t(L1 \text{ Sequencer batch inclusion}) - t(L2 \text{ mapped last block})$

selected sample W_{ingest} 404-751 p50 498 551.32 P_{lag} 7-39

p50 16.5 17.275 25 auto redeem redeem creation L2 block W_{effect}

W_{ingest}

finalized kind=9 L1 event L2 receipt batch L2

block block hash public first-seen Sequencer

Sequencer MEV P_{lag} L2→L1 publication lag L1→L2 message

25 auto redeem redeem redeem

6.6

717

V0/V1

MEV



7.1

L1→L2 message visibility L2→L1 batch publication lag

“ ”

Sequencer

searcher

Commit-Reveal

payload

“

MEV”

7.2

public first-seen Sequencer block timestamp

SDK retryable

Commit-Reveal

“ ”

V0→V1 Nitro SDK L2 /

raw processed archive/RPC ID

receipt V2

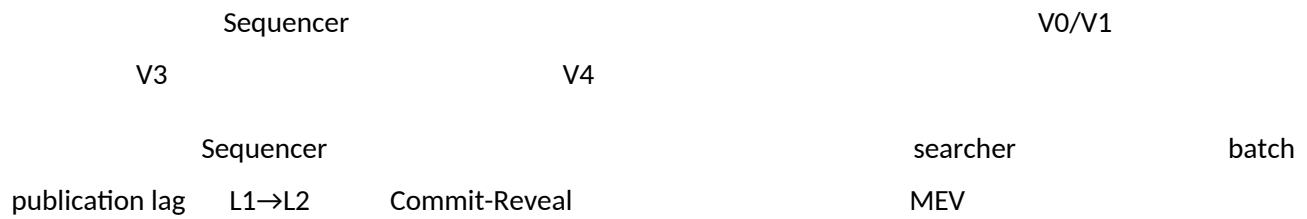
V0 V1 “ / ” V3

V4 Commit-Reveal



RQ1-RQ3 V0-V4

Commit-Reveal Arbitrum



□ □ □ □ □ □ □ □ □ □ □

1. [ppt/](#) Hybrid TEE-Rollup MEV Rollup
2. [_20260717/](#) /
3. [_20260717/](#) / V0-V4 Commit-Reveal
4. [MEV](#) [.md](#) L2 MEV [.md](#) Timeboost
5. Flash Boys 2.0 Quantifying BEV Theory of MEV I MEV Mitigation in Ethereum L2 Arbitrum Timeboost SoK: The Evolution of MEV, From Miners to Cross-Chain SoK: Fair Message Ordering